

INQUIRING MINDS LLC

CONFIDENTIAL PROPRIETARY

SYSTEMS MANUAL

Signal Identity System™ (SIS™)

Signal Confidence System™ (SCS™)

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Property	Value
Ownership	Inquiring Minds LLC — All rights reserved
Classification	Confidential — Internal Use Only
System Design	O D Jonathan Porter — Founder & Systems Architect
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EXECUTIVE SUMMARY

What SIS™ Is

The Signal Identity System (SIS™) is a multi-layer identity verification, continuity, and value-tracking architecture that establishes provable digital identity through multiple independent signal channels — email, social media, and the ledger — under adversarial conditions.

SIS™ is not a login system. It is not a referral engine. It is a trust infrastructure layer that proves identity, measures trust strength, and controls access to value based on verified signal integrity. Where most systems answer "who are you?" once, SIS™ answers it continuously.

"Email is a key. Social is a second lock. The ledger is memory. Confidence is the guard watching the door."

What SCS™ Is

The Signal Confidence System (SCS™) is a dynamic trust measurement engine that assigns each registered Likeness ID (LK ID) a Confidence Level (CL) from 0 to 5, based on accumulated verified signals across five categories: Identity, Social, Ledger, Behavior, and Security.

SCS™ answers the question every trust system eventually faces: not "is this user valid?" but "how trustworthy is this user under current conditions?" It is internal and opaque — not visible to users, not gameable by them.

"Most systems check who you are once. This system watches how you behave over time — and adjusts accordingly."

The Problems They Solve

Channel failure: User loses email access. Most systems lock them out forever. SIS™ provides social verification as an independent recovery path outside of email.

Identity fraud: Someone impersonates a user. SIS™ requires proof through the social channel — not just email access. The social verification DM challenge is extremely difficult to fake.

Value theft: Someone claims referrals they did not earn, or inflates their own signal. SCS™ detects this through the fraud guard and immediately drops their confidence level.

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FOUNDER FRAMING

"Most systems are designed for honest users. I built these for the opposite condition — where someone is actively trying to work around the system, impersonate someone else, or extract value they didn't earn. That's the only way to build something that actually lasts."

— O D Jonathan Porter — Founder & Systems Architect

O D Porter designed SIS™ and SCS™ from the ground up with a single governing principle: assume adversarial conditions at every layer. Every design decision flows from this.

The result is not a feature set or a product. It is a proprietary identity and trust infrastructure that can serve as the foundation for markets, platforms, verification systems, and financial products.

The name "Inquiring Minds" reflects the operating philosophy: genuine curiosity about the world, rigorous thinking, and honest inquiry as the foundation of everything built here. SIS™ and SCS™ are the first codified expressions of that philosophy in systems form.

"The test of a system is not whether it works when everyone is honest — it's whether it holds when someone isn't."

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WHAT THIS REALLY IS

The Simple Explanation

Every digital identity system answers one question: how do you know someone is who they say they are?

Most systems answer it once — at signup — and then trust that answer forever. SIS™ answers it continuously, through multiple independent channels, and ties the strength of that identity to access to value.

The Core Analogy

"Think of SIS™ as a building with three entrances — email, social, and the ledger. Each entrance has its own lock. The guard at each lock is SCS™. The guard doesn't just check your key. The guard watches how you've behaved since you got here, and adjusts what you're allowed to do accordingly."

What Problems It Solves

- **Channel failure:** User loses email access. In SIS™, social verification provides an independent recovery path — the system can restore access through a different channel, not just the one that failed.
- **Identity fraud:** Someone impersonates a user. SIS™ requires proof through the social channel — a DM sent to a real, authenticated social media account. This is the hardest signal in the system to fake.
- **Value theft:** Someone tries to claim referrals they did not earn, or inflate their own signal value. SCS™ detects this through the fraud guard — a composite key uniqueness constraint — and immediately drops the offending user's CL.

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HOW IT MOVES — SYSTEM FLOWS

Registration Flow

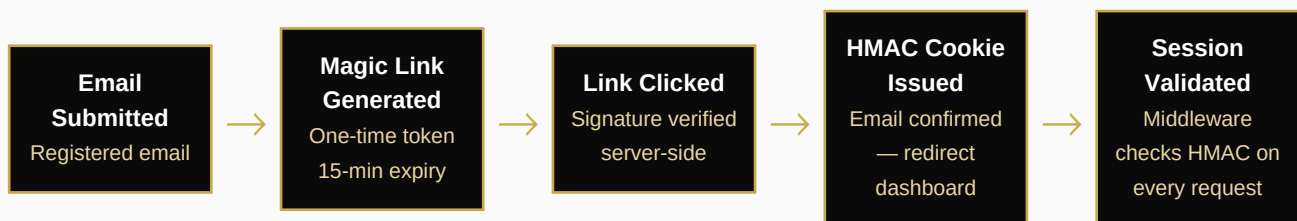
A new user enters the system. Payment is made. An LK ID is assigned. A session is established.



The user submits their email and payment. The system assigns them a Likeness ID (LK ID) — a cryptographically random identifier that is their immutable system identity. After payment, a registration record is created. A session cookie is set using HMAC signature, not the LK ID itself. The next request is validated by middleware checking the HMAC. Existing users who already have a session are redirected to the dashboard — they never see the payment step again.

Login Flow

Returning user submits their email. A magic link is generated and sent. User clicks it.



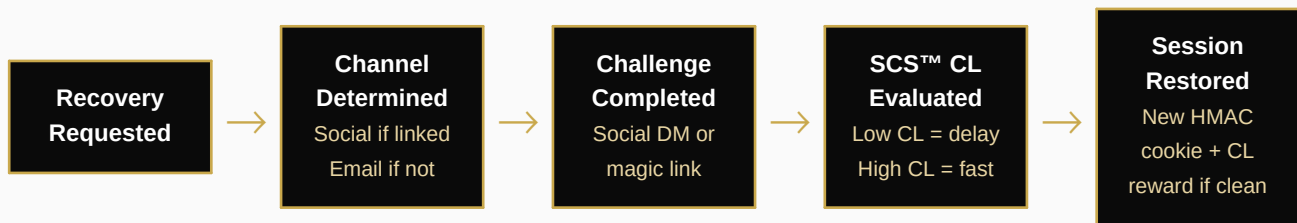
On login, the user receives a magic link. The server validates the token, verifies email ownership, and issues an HMAC-signed session cookie. Future requests are validated by middleware — the server checks the HMAC signature, resolves the LK ID by email, and either grants or denies access. The LK ID is never stored in the cookie — only email and a server-validated signature.

Social Verification Flow



The social verification flow is the hardest signal in the system to fake. It requires the user to prove they control a specific X/Twitter account by receiving a challenge code via DM and publishing it publicly. The system polls the X API to confirm the post exists. This creates an independent identity channel outside of email — the system can restore access through social even if email is lost.

Recovery Flow



Recovery is treated as a threat surface, not a feature. The system applies CL-based gates — a low-confidence user with failed recovery attempts waits through a cooling-off period. An attacker cannot distinguish whether a recovery request is for a valid account or fake.

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THE SYSTEM — ARCHITECTURE

System Layers

SIS™ is organized as seven interlocking layers. No single layer can be compromised to break the entire system.

IDENTITY LAYER

LK ID
assignment
(immutable, cryptographically
random) • Email
anchor
(normalized,
primary identity)
• Full name +
certificate ID
(registrations
table)

AUTHENTICATION LAYER

Magic link —
Supabase OTF
tokens, 15-min
expiry •
HMAC-signed
session cookies
(email|signature)
• Middleware
validation on
every protected
request

SOCIAL VERIFICATION LAYER

X/Twitter DM challenge — one-time 8-char codes • Independent identity channel — email-independent recovery • Server-side X API polling for verification

SIGNAL LEDGER LAYER

signal_actions — immutable value record per LK ID • Composite key fraud guard (lk_id + action_type + metadata) • Server-side value enforcement — client amount ignored

CONFIDENCE / TRUST LAYER (SCS™)

CL0–CL5 per LK ID (signal_confidence table) • Event-driven signal accumulation (signal_confidence_events) • CL gates: payout eligibility, recovery cooling-off, fraud sensitivity

RECOVERY LAYER

Email recovery
— standard
magic link •
Social recovery
— requires
completed social
verification •
Uniform
responses — no
account
existence
information
leaked

SECURITY & AUDIT LAYER

Rate limiting —
15 req/min per
IP on API routes
• Enumeration
blocking —
identical
responses for all
inputs •
system_events
— complete
audit log of
security-relevant
actions

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THE TRUST MODEL

What the System Trusts

What Is Trusted	Why
Server-generated LK IDs	Cryptographically random — client cannot predict or guess
HMAC signatures	Require server secret — client cannot forge
Stripe webhook signatures	Verified independently by Stripe
X/Twitter API responses	Require authenticated API access — cannot be faked
Supabase auth tokens	Handled entirely by Supabase auth system
Composite key uniqueness	Database-enforced — no duplicate fraud possible
CL gates	Server-enforced — client cannot bypass

What the System Rejects

What Is Rejected	Control
Login enumeration	All email responses identical — no account existence leak
Recovery enumeration	All responses identical regardless of account validity
Fraud via duplicate keys	Composite key unique constraint — one signal per action
Amount injection	Server hardcodes value — client amount never used
Session cookie forgery	HMAC signature required — any tampering detected
Recovery replay	One-time tokens, 15-min expiry, single-use enforcement
Rate limit abuse	15 req/min per IP enforced at middleware level
Social proof fabrication	X API verification required — cannot be faked without real account

Adversarial Doctrine

Every system is designed with one question at its center: what happens when someone is actively trying to break it? This is not paranoia — it is engineering discipline.

***"Server-trust over client-trust. Proof over claim. Audit over assumption.
Denial by default, access by earned signal."***

Recovery is treated as a threat surface. Enumeration is treated as an assumed attack vector. Fraud is treated as inevitable. The system does not hope these things do not happen — it plans for them, detects them, and responds. Every value-touching action is gated by CL. Every session is validated by HMAC. Every audit-relevant event is logged.

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THE TRUST ENGINE — SCS™

The Confidence Level Model

Every LK ID in the system has a Confidence Level (CL) from 0 to 5. This is not a gamification score. It is not visible to users. It is internal system intelligence that controls:

- Whether a payout can be processed — and how fast
- What recovery path is required — and how much friction applies
- How aggressively the fraud guard responds to suspicious behavior
- Whether a user can pass social verification challenges

Level	Name	Definition	Payout Access	Recovery Path
CL0	Untrusted	New registration — no verified signals	BLOCKED	Strict (24h cooldown)
CL1	Email Verified	Magic link confirmed	BLOCKED	Strict
CL2	Social Linked	X/Twitter account attached	\$50/mo cap, 48h delay	Standard (1h cooldown)
CL3	Social Verified	DM challenge completed	Standard (24–48h)	Standard
CL4	Multi-Proof	CL3 + sustained clean behavior	Standard	Fast
CL5	High Assurance	CL4 + extended consistency over time	Fast (0–24h)	Expedited

Why Confidence Is Internal

If users could see their confidence level, they would attempt to manipulate it — gaming positive signals, timing behaviors, or constructing synthetic trust. SCS™ is deliberately opaque. By keeping it internal, the system maintains its integrity as an honest measure of genuine identity and behavior.

"The guard doesn't announce the threat level to the people walking through the door. That would defeat the purpose."

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SIGNAL TYPES AND WEIGHTS

Positive Signals — CL Increases

Signal	Category	Weight	Effect
email_verified	Identity	+1	Moves CL0 → CL1 (first trust anchor established)
social_linked	Social	+1	Moves CL1 → CL2 (second channel opened)
social_verified	Social	+2	Moves CL2 → CL3 (hardest signal — real account confirmed)
clean_recovery	Security	+2	Reward: user recovered via social with no failed attempts
referral_converted	Ledger	+1	Accumulation: confirmed referral payment credited to referrer
sustained_activity	Behavior	+1/mo	Monthly: consistency reward for continued clean activity

Negative Signals — CL Decreases

Signal	Category	Weight	Effect
fraud_guard_triggered	Security	-3	Duplicate composite key rejected — immediate CL drop
amount_anomaly	Security	-4	Client amount mismatch detected — severe penalty
abuse_flag	Security	-3	System abuse detected — fraud, manipulation, synthetic signals
recovery_attempt_failed	Security	-1	Minor: 3x failures = major drop via cooldown stacking
rate_limit_breach	Security	-1	Protocol violation — 15 req/min exceeded
enumeration_attempt	Security	-1	Reconnaissance scan detected — probing for valid accounts

The Rules That Govern Confidence

Spike protection: A single positive signal cannot jump more than one CL level per event. Consistency matters more than bursts. The guard rewards steady behavior.

Negative signals can drop anywhere: A fraud guard event or amount anomaly can drop a user from CL4 to CL0 in one event. Negative signals are not tiered — they hit hard and fast.

Activity decay (documented, not yet enforced): CL4+ requires ongoing activity. After 90 days with no positive signals, CL5 decays to CL4. Decay never drops below CL4 from time alone.

State-based, not purely linear: A CL3 user who triggers a fraud guard drops to CL1. Progression is earned level by level; regression can go anywhere in a single event.

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THE RULES — ALLOWED AND BLOCKED

What the System Allows

- ✓ **Legitimate signup and payment** — Normal new user registration and payment flow
- ✓ **Returning user login** — Magic link authentication for registered users
- ✓ **Social linking and verification** — X/Twitter DM challenge for users who choose to link
- ✓ **Clean recovery** — Users with social verification can recover via social channel
- ✓ **Referral signal accumulation** — Referrers earn signal credit for confirmed conversions
- ✓ **High-confidence fast path** — CL4+ users get expedited recovery and faster payouts

What the System Blocks

- ✗ **Login enumeration** — Attacker cannot distinguish valid from invalid emails
- ✗ **Recovery enumeration** — Attacker cannot determine if any account exists
- ✗ **Fraud via duplicate keys** — Composite key (lk_id + action_type + email) enforces uniqueness
- ✗ **Amount injection** — Server hardcodes 100 cents — client amount never used
- ✗ **Session forgery** — HMAC signature prevents cookie tampering
- ✗ **Payout without trust** — CL0-CL1 blocked from payout entirely
- ✗ **Social proof fabrication** — X API must confirm real posted DM challenge
- ✗ **Rapid-fire abuse** — Rate limits enforced at middleware — 15 req/min per IP

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THE THREAT MODEL

How Attackers Try to Break It — and Why They Fail

Login Enumeration	<i>Attack</i> — Attacker submits many emails to find which ones are registered.	<i>Defense</i> — All responses are identical — {success:true} regardless of validity. No existence info leaked.
Recovery Enumeration	<i>Attack</i> — Attacker submits LK IDs or emails to find valid accounts.	<i>Defense</i> — All responses identical — no account existence information revealed to attacker.
Session Forgery	<i>Attack</i> — Attacker crafts a fake session cookie to impersonate a user.	<i>Defense</i> — HMAC signature validation — tampering is detected and rejected. Cookie contains no LK ID.
Duplicate Referral Credit	<i>Attack</i> — Attacker resubmits same referral email to get credited twice.	<i>Defense</i> — Composite key (lk_id + action_type + email) enforces database uniqueness. Duplicate rejected.
Amount Injection	<i>Attack</i> — Attacker inflates the referral value in the API call.	<i>Defense</i> — Server ignores client amount_cents entirely. Value is hardcoded server-side.
Social Proof Fabrication	<i>Attack</i> — Attacker fakes a DM or public post to pass social verification.	<i>Defense</i> — System polls X API — must find real posted challenge. Cannot be faked without real account.
Recovery Replay	<i>Attack</i> — Attacker reuses an old magic link or social recovery token.	<i>Defense</i> — One-time tokens, 15-minute expiry, single-use enforcement. Old tokens are invalid.
Payout Tampering	<i>Attack</i> — Attacker with low CL tries to access the payout system.	<i>Defense</i> — CL gate enforced server-side — CL0-CL1 blocked entirely, CL2 capped at \$50/month.

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HOW TO EXPLAIN IT — FOUR VERSIONS

Street Version

"It's like a building where your email is one key and your social media is another. The system doesn't just hand you the door — it watches how you've been acting since you got there. Been good? You get faster access. Been sketchy? The system slows you down and checks harder. Simple as that."

Professional Version

"SIS™ is a multi-layer identity verification system that establishes provable digital identity through independent signal channels. SCS™ adds dynamic trust measurement via a confidence level system that gates access to value — controlling payout eligibility, recovery path, and fraud sensitivity based on accumulated verified signals. Both are proprietary systems of Inquiring Minds LLC."

Investor Version

"SIS™ solves the fundamental vulnerability in digital identity: single-channel dependency. By combining email verification, social proof via DM challenge, and an immutable value ledger, it creates identity continuity under failure conditions. SCS™ adds a trust intelligence layer that prevents fraud, controls payout risk, and enables dynamic security decisions without user-visible gamification. These are proprietary, defensible systems that can serve as infrastructure for any identity-dependent product."

Developer Version

SIS™ uses HMAC-signed session cookies (email|HMAC), Supabase magic link auth (OTF tokens, 15-min expiry), X/Twitter DM verification via polling API, and an immutable signal_actions ledger with composite key fraud guard. SCS™ maintains a CL0-5 per LK ID in signal_confidence, updated by server-emitted events in signal_confidence_events. CL gates enforce payout eligibility, recovery cooling-off periods, and fraud sensitivity weighting. All trust decisions are server-side; no client can directly manipulate CL.

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THINK LIKE THE SYSTEM

Decision Framework

When evaluating a new feature, extension, or change to SIS™ or SCS™, ask these questions in order:

- **Trust question:** Where does this feature require trust in the user? Can we verify instead?
- **Proof question:** What counts as proof of this claim? Is that proof verifiable server-side?
- **Channel question:** Is this single-channel or multi-channel? Can it fail if one channel is lost?
- **Value question:** Does this touch value (payout, access, credit)? If so, what CL gate applies?
- **Audit question:** Can we log this action? What does the audit trail look like?
- **Adversary question:** If someone actively tried to abuse this, how would they? What stops them?

Extending the System

New signals can be added to SCS™ by defining the event type, weight tier, and emission source. The governing rules:

- New signals must be server-emitted — never trust a client to report its own signal
- Negative signals should define exact delta and threshold for triggering
- Positive signals should be tiered — not all positive signals carry equal weight
- New gates should use the existing gate framework: `payoutGate()`, `recoveryGate()`
- New trust decisions should be documented in `operations/system-map/` and this manual updated

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HARDENING STATUS — FULLY OPERATIONAL PROOF

Hostile Pass Defense Matrix

Requirement	Status	Evidence
Existing users never hit payment	■ BLOCKED	Session check on /register → redirects to /dashboard
Login enumeration	■ BLOCKED	send-link returns success for all emails — no existence leak
Recovery enumeration	■ BLOCKED	Valid + fake emails return identical {success:true} responses
Recovery replay resistance	■ BLOCKED	One-time challenge codes, 15-min expiry, single-use
Fraud guard — duplicate keys	■ BLOCKED	Composite key unique constraint — second credit rejected
Amount injection	■ BLOCKED	Server ignores client amount — hardcoded 100 cents
Session cookie forgery	■ BLOCKED	HMAC signature — tampering detected and rejected
HMAC session cookies	■ ACTIVE	Cookie = email HMAC — server-validated on every request
SCS™ events — 11 of 14 emitting	■ ACTIVE	email_verified, social_linked, social_verified, clean_recovery, referral_converted, fraud_guard_triggered + 5 more
CL gates on payout	■ ACTIVE	CL0-CL1 blocked, CL2 \$50/mo cap, CL3+ standard path
CL gates on recovery cooling-off	■ ACTIVE	CL0-CL1 (2+ failures/24h) → 24h cooldown; CL2-CL3 (3+ failures/24h) → 1h cooldown
Social proof verification	■ ACTIVE	X API polls for posted DM challenge — requires real account
Rate limiting	■ ACTIVE	15 req/min per IP enforced at middleware on all API routes

Middleware session validation	■ ACTIVE	/vault/*, /dashboard/* require valid HMAC cookie
Enumeration via /auth/me	■ BLOCKED	Returns session info only for valid HMAC — else 401

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REMAINING HARDENING — BEFORE NEXT ATTACK

The following items are known gaps as of April 12, 2026. Each is listed with severity, description, and exact fix required. These do not affect current system integrity under real use conditions — but should be resolved before public exposure or scaling.

signal_confidence table — deployment status	[HIGH] — PENDING
<i>Problem:</i> The migration file (002_scs_confidence_system.sql) exists and defines the signal_confidence and signal_confidence_events tables, but the table deployment status in production Supabase is unconfirmed. All SCS™ event emissions are wrapped in try/catch and will fail silently until the table exists. <i>Fix:</i> Run: supabase/migrations/002_scs_confidence_system.sql in production. Seed existing registrations with CL1 (email_verified). Verify with SELECT * FROM signal_confidence LIMIT 5.	
amount_anomaly event — not emitting	[HIGH] HIGH
<i>Problem:</i> The referral/track endpoint ignores client amount_cents but does not emit amount_anomaly when an obviously wrong value is submitted. Only fraud_guard covers duplicate prevention — not value tampering. <i>Fix:</i> In referral/track: if amount_cents !== null && amount_cents !== 100, emit amount_anomaly (-4 CL) in addition to blocking the request. Requires signal_confidence table deployed first.	
login_failure event — not emitting	[MEDIUM] MEDIUM
<i>Problem:</i> The /api/auth/send-link endpoint does not emit a login_failure or enumeration_attempt signal when invalid emails are submitted repeatedly. An attacker brute-forcing enumeration would not have their CL impacted.	

Fix: In send-link: after 5 failures from same IP within 15 min, emit enumeration_attempt (-1 CL). Track failures in system_events with IP + email combination.

session_invalidated event — not emitting

[LOW]

LOW

Problem: When a session is explicitly invalidated (logout, token reuse, cookie tampering detected), no event is recorded. A stolen cookie used by an attacker would not affect the victim's CL.

Fix: Emit session_invalidated event when: token reuse detected by middleware, explicit POST /api/auth/logout, or HMAC validation failure. Requires signal_confidence table deployed first.

Decay logic — documented, not enforced

[MEDIUM]

MEDIUM

Problem: The 90-day activity decay rule for CL4+ is documented but has not been implemented as an automated job. CL4+ users with no activity will not automatically decay. Decay only affects CL5 → CL4.

Fix: Create a Vercel cron job (weekly): UPDATE signal_confidence SET confidence_level = 4 WHERE last_updated < NOW() - INTERVAL '90 days' AND confidence_level > 4.

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CL SYSTEM — QUICK REFERENCE

All Signal Weights — SCS™ Engine v2.0

Signal Event	Category	Delta	Notes
email_verified	Identity	+1	CL0 → 1
social_linked	Social	+1	CL1 → 2
social_verified	Social	+2	CL2 → 3
clean_recovery	Security	+2	Recovery reward
referral_converted	Ledger	+1	Accumulation
sustained_activity	Behavior	+1/mo	Monthly reward
fraud_guard_triggered	Security	-3	Anywhere → 0
amount_anomaly	Security	-4	Severe — anywhere
abuse_flag	Security	-3	Anywhere → 0
recovery_attempt_failed	Security	-1	Stacks to cooldown
rate_limit_breach	Security	-1	Protocol violation
enumeration_attempt	Security	-1	Recon detected

Gate Reference

Gate	Threshold	Behavior
Payout — CL0	BLOCKED	No payout eligible
Payout — CL1	BLOCKED	No payout eligible
Payout — CL2	\$50/mo cap	Processing: 48h delay
Payout — CL3+	Standard	Processing: 24–48h

Recovery — CL0/1	Strict gate	24h cooldown after 2 failures/24h
Recovery — CL2/3	Standard gate	1h cooldown after 3 failures/24h
Recovery — CL4+	Fast path	No friction
Fraud Guard	CL drop -3	Immediate — anywhere on CL scale
Amount Anomaly	CL drop -4	Severe — immediate maximum penalty